

# Sohini Banerjee

(+91) 98740 38011 | [son20apakhi05@gmail.com](mailto:son20apakhi05@gmail.com) | [linkedin.com/in/sosush](https://linkedin.com/in/sosush) | [github.com/sosush](https://github.com/sosush)  
[sohinibanerjee.vercel.app](https://sohinibanerjee.vercel.app) | [leetcode.com/u/sb\\_1315](https://leetcode.com/u/sb_1315)

## EDUCATION

### Vellore Institute of Technology (VIT)

Chennai, TN

*Bachelor of Technology in Computer Science and Engineering (AI & ML)*

*Jul. 2024 – Present*

- CGPA: 8.73 / 10.0
- Coursework: Data Structures and Algorithms, Operating Systems, DBMS, Computer Networks, OOP.

### Delhi Public School

Kolkata, India

*Senior Secondary (Class XII) - Science*

*Mar. 2022 – Mar. 2024*

- Score: 91.2%

## EXPERIENCE

### Software Development Intern

Jun. 2024 – Aug. 2024

*Hackfinity*

*Chennai, India*

- Developed and deployed responsive web interfaces using **React.js** and **JavaScript**, reducing page load time by 25%.
- Collaborated with the backend team to integrate **RESTful APIs**, ensuring seamless data fetching for 1000+ daily users.
- Fixed critical bugs in the frontend codebase and documented technical specifications to support future development.

## PROJECTS

### PRISM Protocol | *Java, Spring Boot, Solidity, OpenCV, FastAPI*

Oct. 2024

- Designed a decentralized identity system using **Spring Boot microservices** for deepfake detection and secure authentication.
- Implemented real-time liveness detection using **OpenCV**, achieving 99% accuracy against AI replay attacks.
- Deployed Smart Contracts on **Polygon** using **Solidity** to issue Soulbound Tokens (SBTs) for immutable identity verification.

### Spark | *Next.js, TypeScript, Framer Motion, Tailwind CSS*

Nov. 2024

- Built a high-fidelity "vibecoding" application focusing on aesthetic-driven UI and seamless data-to-design transitions.
- Integrated **Framer Motion** and **Magic UI** to create interactive, fluid components for an enhanced user experience.
- Leveraged **Next.js App Router** and **TypeScript** to ensure type-safety and optimized server-side rendering.

### ML-IDS (Intrusion Detection) | *Python, Scikit-Learn, Pandas*

Aug. 2024

- Built a cybersecurity security model using **Random Forest** and **SVM** classifiers to detect DoS and Probe network attacks.
- Engineered data pipelines using **Pandas** and **SMOTE** for class balancing, achieving a 99.2% detection accuracy.
- Reduced false positive rates to under 0.5% through advanced feature selection and hyperparameter tuning.

## TECHNICAL SKILLS

**Languages:** Java, Python, TypeScript, JavaScript, SQL, C++, Solidity, HTML/CSS

**Frontend:** React.js, Next.js, Framer Motion, Tailwind CSS, Shadcn UI, Three.js

**Backend/AI:** Node.js, Spring Boot, FastAPI, PostgreSQL, Supabase, Scikit-Learn, TensorFlow, PyTorch

**Tools/Ops:** Git, Docker, Linux, AWS, Postman, VS Code, MySQL

**Concepts:** Data Structures, Algorithms, System Design, Blockchain Security, Genetic Algorithms